

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I Shanmuganathan
I 192511397 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Sgt
Signature of the student:

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Q1 : Smart Homes Automation System;

An intelligent agent is a System that perceives its environment through sensor and takes actions using actuators to achieve specific goals.

1) Simple Reflex Agent:-

- * Acts only on the current input using predefined rules.

- * Eg:- Turns on lights when motion is detected and turns them off when no motion detected.

- * Advantage : Fast and Simple

- * Disadvantage:-

Cannot remember past events or learn user preference

2) Model Based Agents

- * maintain an internal model of the environment

- * Example:- Remembers that the user usually return home at 6 Pm and prepares the home accordingly.

* Advantage:-

Handles partially observable environments.

* Disadvantages:-

Requires more memory and processing

3) Goal Based Agent:-

* Chooses actions to achieve a specific goal

* Example :- Maintains a comfortable room temperature while reducing unnecessary energy usage.

* Advantage : Flexible and plans actions

* Disadvantages : Planning may take additional time

4) Utility Based Agent:-

* Mechanism : Uses a utility function to evaluate how well a goal is achieved when trade off exist.

* Functions:- Weights competing objectives such as user comfort vs. electricity cost. It might cool the house slightly less during peak electricity tariff hours to max financial utility while preserving acceptable comfort.

Q2 Robot in an unknown Maze

When a robot is placed in an unknown maze, it must search for the exit without knowing the path. Uniformed search Algorithm help the robot explore the maze systematically until it reaches the goal.

Uniform Cost Search (UCS):

Uniform Cost Search expands the path with the lowest total cost first. It guarantees the optimal path if all cost are positive. However, it requires a large amount of memory and may take more time when the search space is very large.

Iterative Deepening Search (IDS):

Iterative Deepening Search repeatedly performs depth limited searches with increasing depth limits. Required less memory than UCS and is suitable the depth of the solution is unknown. However repeat the search at each depth level increases computation time.

Relation of intelligent Agents:-

A simple reflex agent cannot solve the maze effectively because it has no memory. A model based agent remembers visited locations and avoids repeated exploration. A goal based agent searches specifically for the exit while a utility based agent selects the safest (or) least expensive path.

Justification:-

A Goal Based Agent with uniform cost search is the most suitable combination because the goal based agent focuses on reaching the exit. While UCS guarantees the lowest-cost path. If memory is limited IDS can be used as an alternative.

Q3 Smart Traffic management System:-

A Smart traffic management system uses AI to control traffic signal based on vehicle density, pedestrian movement, weather conditions, and historic traffic pattern. The aim is to reduce congestion and improve road safety.

Model-Based Agent:-

A Model Based agent stores information about previous traffic conditions and road safety. It uses this knowledge to make more accurate decisions and improve traffic flow.

Goal Based Agent:-

A Goal Based Agent plans traffic signal timing to achieve specific objectives such as reducing waiting time, improve vehicle movement, and ensuring pedestrian safety.

Utility Based Agent:-

A utility Based agent evaluates different traffic control options and selects the one that provides the greatest overall benefit. It balances travel time, fuel consumption, and road safety.

Justification:-

A hybrid model-based and utility based agent is the best choice because it can learn traffic patterns and make intelligent decisions that minimize congestion, reduce travel time, and improve the efficiency of the entire traffic system.